Distillation and Ablation Analysis of Encoder-Decoder Language Models

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Outline

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Introduction

Problem

Encoder-decoder models such as T5 and Flan-T5 are strong instruction-following models, but their internal robustness under compression and structural perturbation is not fully clear.

Goal

We study how distilled encoder-decoder models behave under ablation, masking, and noise, and whether distillation preserves not only output quality but also internal robustness.

Why This Project?

Compression Question

Can smaller distilled models preserve the behavior of larger encoder-decoder teachers?

Robustness Question

Which parts of the model are structurally critical: encoder layers, decoder layers, attention heads, or feed-forward blocks?

Interpretability Question

Do ablations reveal architectural bottlenecks that are hidden by aggregate evaluation scores?

What we try to answer?

1. Does distillation preserve semantic performance under instruction-following evaluation?
2. Does distillation preserve the teacher's internal robustness profile?
3. Are failures concentrated in specific architectural components?
4. Are encoder and decoder layers equally important?
5. Are attention and feed-forward modules equally robust to perturbation?

Outline

What is Model Distillation?

It is a technique where a smaller neural network is trained to replicate the knowledge and performance of a larger, more complex model.

Distillation: Experimental Setup

Methods used:

Semantic Knowledge Distillation: learn to predict the teachers response.
MiniPLM: Filter the data and train only on examples that are easy for the teacher but difficult for the student.

Teacher Model

`Qwen/Qwen3.5-4B`

Student Models

- `google/flan-t5-large`
- `google-t5/t5-large`

Distillation: Results

Dataset	Distillation Method	T5-Large	Flan-T5-Large
Dolly	Semantic	**0.8341**	**0.8754**
	MiniPLM	0.6651	0.5489
Self Instruct	Semantic	**0.8232**	**0.8276**
	MiniPLM	0.7235	0.6183

Table: Embedding similarity scores for Semantic KD and MiniPLM distillation, evaluated with Qwen3.5-0.8B.

Outline

Ablation Setup

Models and Datasets

- **T5-Large**: evaluated on *Dolly-15k*
- **Flan-T5-Large**: evaluated on *Self-Instruct*
- Each model is compared against two distilled variants:
  - Semantic Knowledge Distillation student
  - MiniPLM student

Ablation Families

- **Block dropping**: single, cumulative, odd/even encoder-decoder block removal
- **Feed-forward interventions**: structured FF dropping and random FF masking
- **Attention masking**: encoder self-attention, decoder self-attention, decoder cross-attention
- **Gaussian noise**: perturbation of block and feed-forward representations

Outline

T5

Model and Dataset

- Base model: `t5-large`
- Dataset: *Dolly-15k*
- Evaluation subset: 128 examples
- Decoding: greedy decoding, `num_beams=1`

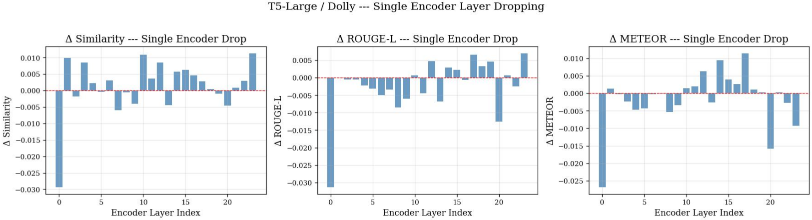
Ablation Families

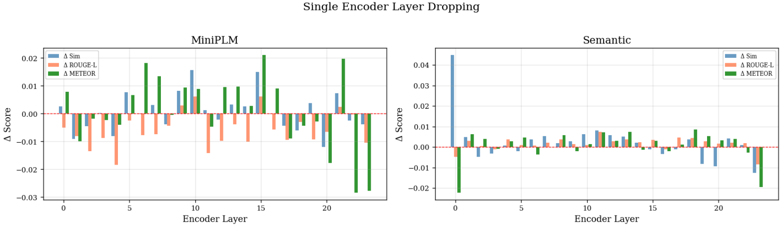
- Encoder and decoder block dropping
- Odd/even block and feed-forward dropping
- Attention and feed-forward masking
- Gaussian noise injection

Metrics

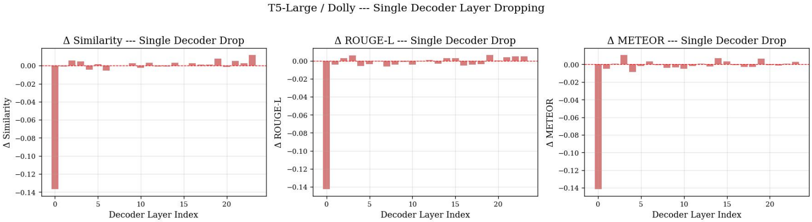
Similarity, ROUGE-L and METEOR.

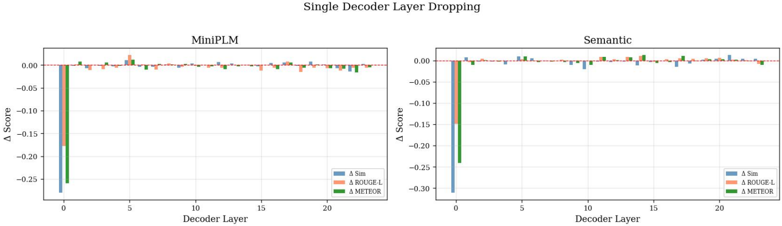
T5 Base vs. Students: Single Encoder Block Dropping



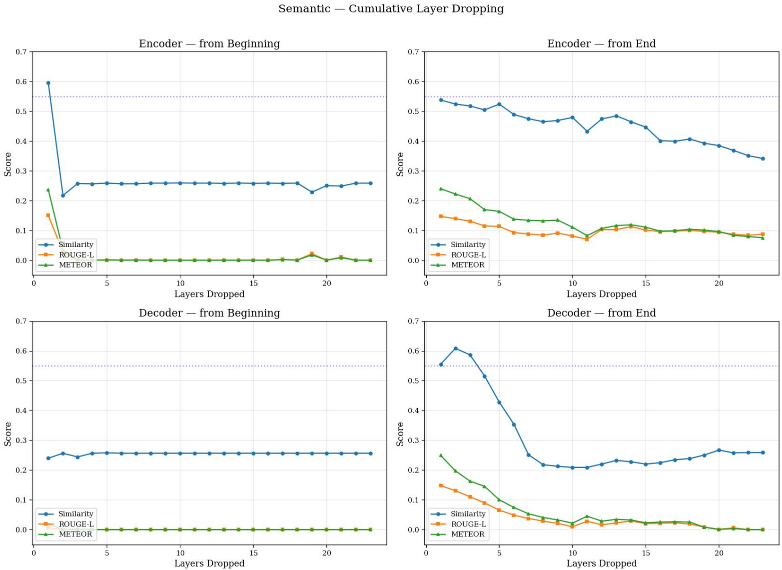


T5 Base vs. Students: Single Decoder Block Dropping

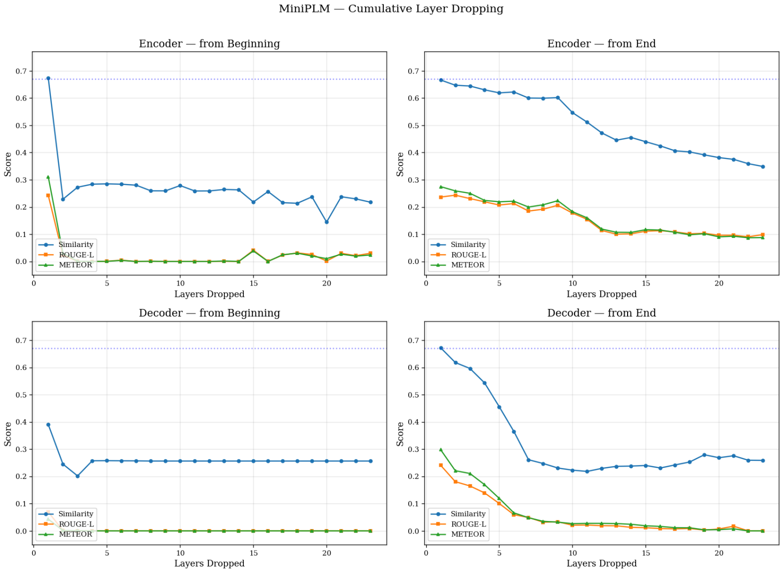




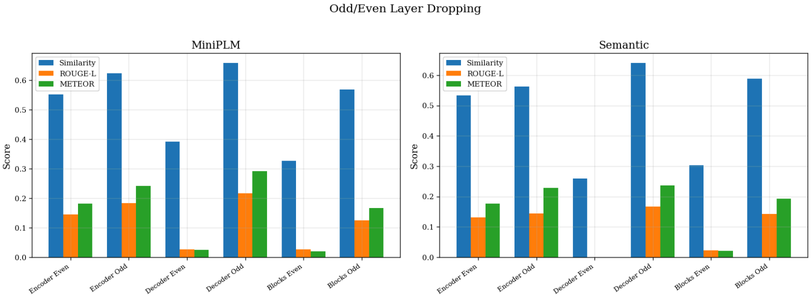
T5 Semantic Student: Cumulative Layer Dropping



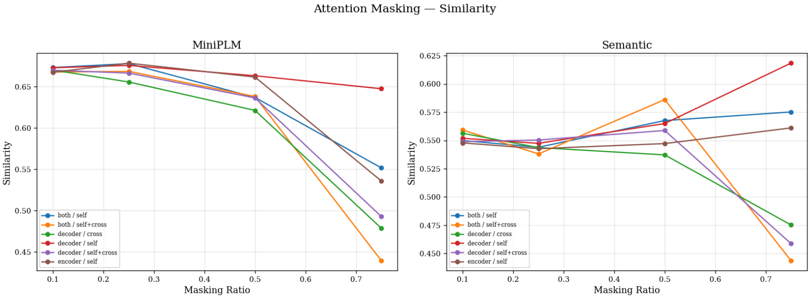
T5 MiniPLM Student: Cumulative Layer Dropping



T5 Students: Odd/Even Layer Dropping



T5 Students: Attention Masking



T5 Ablation: Summary

- Dropping first Encoder/Decoder layer on any architecture results in significant performance loss.
- Cumulative Dropping from the end of the stack is less destructive as starting from the beginning.
- Dropping even layers is more destructive than dropping odd ones.
- Masking both self and cross attention in both encoder and decoder results in largest performance drop.

Outline

1. Introduction
2. Distillation
3. Ablation
4. T5
5. Flan-T5
6. Future Work

Flan-T5

Model and Dataset

- Base model: `google/flan-t5-large`
- Dataset: *Self-Instruct*
- Evaluation subset: 128 examples
- Decoding: greedy decoding, `num_beams=1`

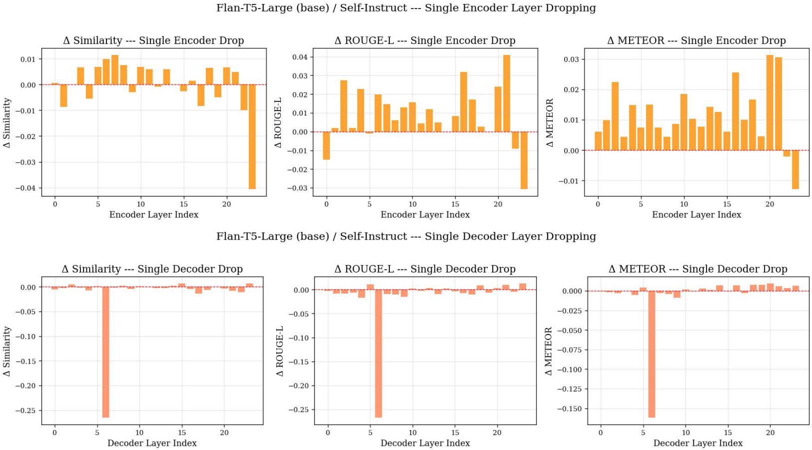
Ablation Families

- Encoder and decoder block dropping
- Odd/even block and feed-forward dropping
- Attention and feed-forward masking
- Gaussian noise injection

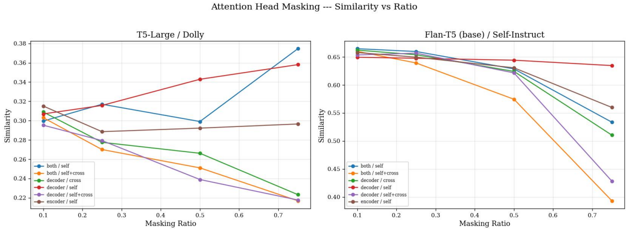
Metrics

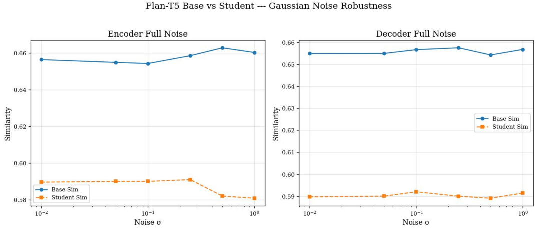
Similarity, hard semantic prediction, ROUGE-L and METEOR.

Flan-T5 Base: Single Block Dropping

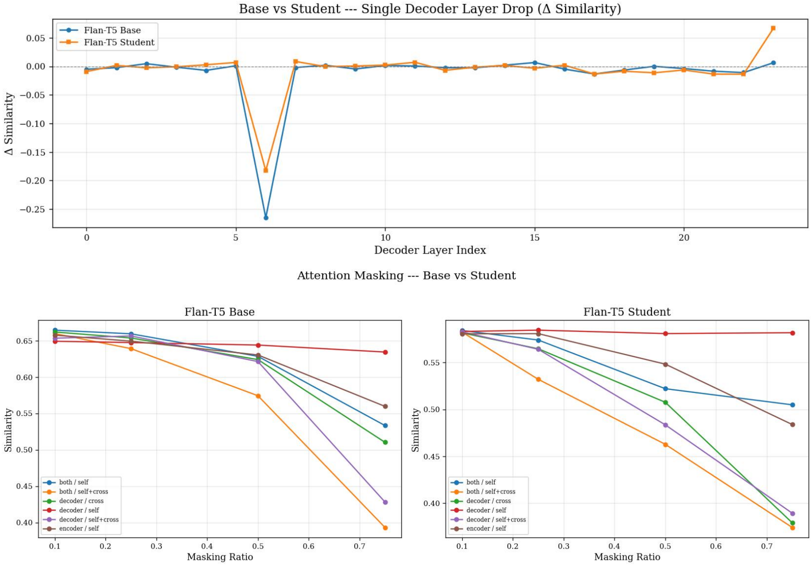


Flan-T5 Base: Masking and Noise

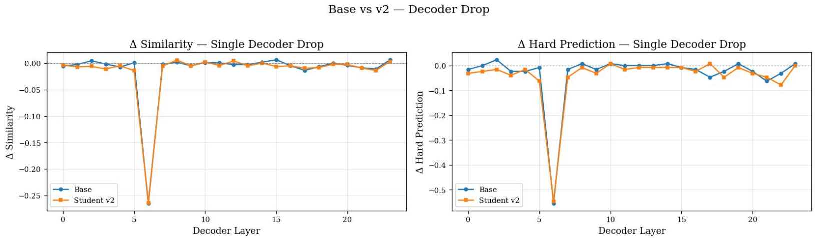


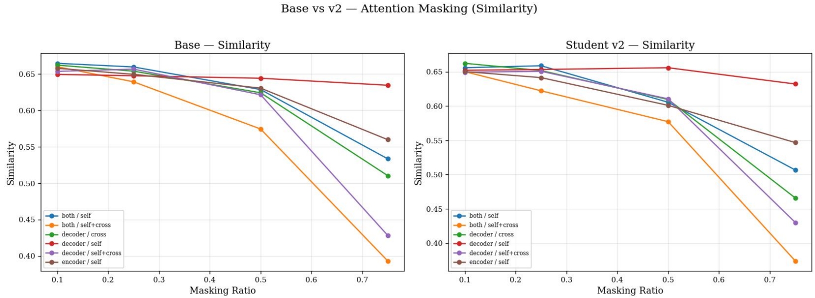


Flan-T5: Base vs Semantic KD Student

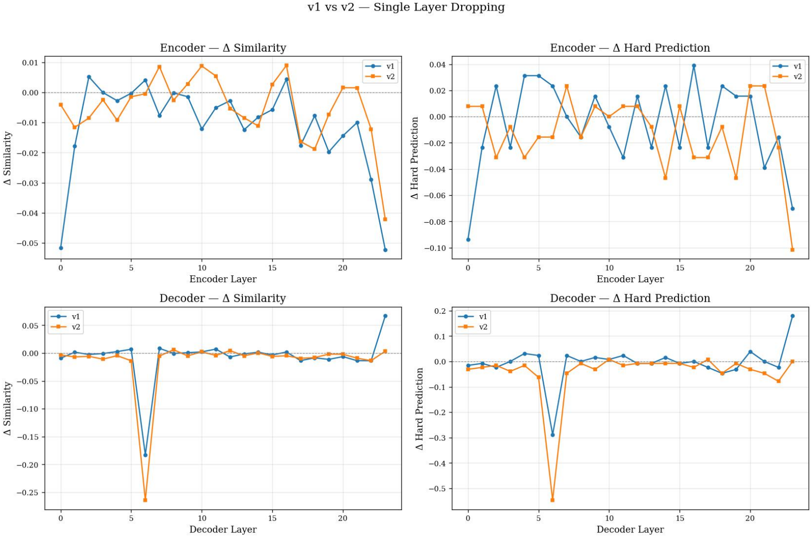


Flan-T5: Base vs MiniPLM Student

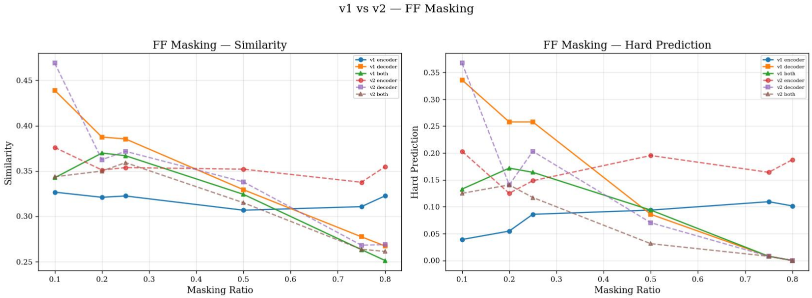




Flan-T5 Ablation: Student v1 vs Student v2



Flan-T5 Ablation: Student v1 vs Student v2



Flan-T5 Ablation: Summary

- Base Flan-T5 has a localized decoder bottleneck.
- Semantic KD changes robustness more aggressively.
- MiniPLM preserves teacher-like behavior more strongly.
- Preserving the teacher also preserves its structural weaknesses.

Outline

Future Work

- Extend the same ablation protocol to additional encoder-decoder families and larger instruction-following datasets.
- Test whether removable blocks and feed-forward sublayers can be permanently pruned, followed by short retraining.
- Move from block-level analysis to finer interventions:
  - attention-head pruning;
  - neuron-level feed-forward analysis;
  - activation patching;
  - representation similarity analysis.
- Evaluate under additional decoding strategies, including beam search and sampling-based decoding.
- Complement embedding and lexical metrics with likelihood-based metrics and human evaluation.

Thank you for your attention!